PHASE 7 OF 8

Testing, Troubleshooting & FAQ

A practical test plan, fixes for the most common problems, and answers to frequently asked questions.

Test Plan | Troubleshooting | FAQ

7.1 Testing Recommendations

The project does not currently ship automated tests. These are the highest-value places to start:

Priority	Area	What to confirm
1	`quiz_module.py` — `clean_json_block()` and validation	The most failure-prone path, since it depends on the model returning well-formed JSON
2	`main.py` input validation	Each POST route returns 400 with a clear message when its field is missing or empty
3	`explanation_module.py` caching	Model and tokenizer load only once across multiple requests
4	Error handling in Gemini modules	Each returns a friendly string, not an exception, when the key is missing or the call fails

Example test sketch (pytest)

```
def test_clean_json_block_strips_fences():
    raw = '```json\n[{"question": "Q"}]\n```'
    assert clean_json_block(raw) == '[{"question": "Q"}]'

def test_generate_quiz_returns_error_without_key(monkeypatch):
    monkeypatch.setattr(quiz_module, "GEMINI_API_KEY", None)
    result = quiz_module.generate_quiz("any text")
    assert result[0]["error"]
```

7.2 Manual Smoke-Test Checklist

Check	Expected result
Open the home page	Q&A, Explain and Summarize forms appear
Submit an empty Explain form	Clear “Please provide a topic.” error (HTTP 400)
Ask “Why is the sky blue?”	A written answer appears under the form
Generate a quiz for “python”	Three questions, four options each; Check Answer gives instant feedback
Run Explain twice	Second call is noticeably faster (model cached)

7.3 Troubleshooting Guide

"GEMINI_API_KEY is not set"

The .env file is missing, misnamed or not being loaded. Confirm it is named exactly `.env`, sits in the project root next to `main.py`, and that `load_dotenv()` runs before the key is read.

"model not found" errors from Gemini

Run `python check_models.py` to list the models your key can actually call, then update the model name in `qna.py`, `summary_module.py`, `quiz_module.py` and `learning_path.py` to match.

Quiz returns a parsing error instead of questions

Gemini occasionally wraps JSON in Markdown fences or adds stray commentary. `clean_json_block()` strips common fence patterns, but a badly formed response can still fail `json.loads()`. Rerunning the request, or tightening the prompt's formatting instructions, usually resolves it.

First Explain request is slow

Expected — the first call downloads `MBZUAI/LaMini-Flan-T5-783M` (a few hundred MB). Later calls use the cached weights and are noticeably faster.

Endpoint hangs with no response

Because the Gemini-calling modules do not set a request timeout, a slow API response can look like a hang rather than an error. Adding a timeout to each Gemini call is the recommended fix.

Symptom	Quick diagnosis	First action
Blank page at /	templates/index.html missing	Verify the templates folder
Unstyled page	static/style.css missing or not mounted	Verify the static folder
400 error	Empty required field	Fill in the field and resubmit
Errors on every Gemini feature	Key or model-name problem	Check .env, then run check_models.py

7.4 Reading HTTP Status Codes

Code	Meaning in EduGenie	What to do
200	Request succeeded	Nothing — the JSON result is in the body
400	A required field was empty	Send a non-empty topic or text value
422	Request did not match the expected format (FastAPI validation)	Check parameter names, e.g. `?question=`
500	Unhandled server error	Read the uvicorn terminal output for the traceback

7.5 Frequently Asked Questions

Does EduGenie work without an internet connection?

Partially. Explain runs a local model and works offline once the weights are cached. Q&A, Summarize, Quiz and Learning Path call the Gemini API and need internet access.

Why does Explain sometimes phrase things more simply than Q&A?

`explanation_module.py` explicitly prompts for an explanation "for a school student" and uses a smaller model tuned for concise output, whereas `qna.py` passes the question through with minimal extra instruction.

Can I change which Gemini model is used?

Yes. The model name (currently `gemini-3.6-flash`) is set inside each Gemini-calling module. Run `check_models.py` first to confirm the replacement is available to your key.

What happens if the quiz JSON is malformed?

The module validates that the parsed result is a list. If parsing fails or the shape is wrong, it returns a friendly error object instead of letting the exception crash the request.

Is my Gemini API key ever exposed to the browser?

No. The key lives in the .env file on the server and is read only by the backend via `os.getenv("GEMINI_API_KEY")`. The browser talks only to the FastAPI routes.

Why cache the model in a dictionary instead of a global variable?

Either works. A small dictionary keeps the tokenizer and model together and makes it easy to check "has this been loaded?" in one place, which is what `_load_model()` does.

Can I add a feature without touching existing routes?

Yes. Add a new module (for example `flashcards_module.py`) with one function, then add a single new route in `main.py` — the same pattern used by `quiz_module.py`.

How do I run the app on a different port?

Pass the port to the server, for example `uvicorn main:app --reload --port 8080`, then open that port in the browser.

Where are the local model weights stored?

Hugging Face caches downloaded weights on disk (by default under your user profile's `.cache/huggingface` folder). Deleting that model's folder forces a fresh download on the next Explain request.

Does EduGenie remember my previous questions?

No. The backend is stateless, so each request is handled on its own. Persisting history is listed as a future enhancement in Phase 8.